Carbon-aware economic viability of Ontario underwater robotics: finite-horizon accounting, quantum kernels and post-quantum report verification

Independent Nature-style research draft | Not a Nature publication or peer-reviewed study

Abstract

We couple an editable ten-year fleet cost and carbon-footprint scenario with a finite product-state kernel demonstrator and local post-quantum report verification. The model separates discounted financial value, carbon shadow value and lifecycle emissions. Arithmetic scheduling labels reuse a helper from Quantum Privacy Cloak, but its simulated security measures are not relied upon. Real ML-DSA-65 and ML-KEM-768 functional tests exercise signatures, tampering and shared-secret agreement. All economic inputs are assumptions: no field-calibrated lifecycle assessment, quantum advantage, carbon-credit eligibility or production security is established.

1. Scope, assumptions and reproducibility

The comparison is an incremental new robotic fleet against incumbent conventional monitoring. Robot and QML facility embodied emissions are charged at year zero. The baseline's existing assets are assumed sunk; its annual carbon inventory is user-entered. Both alternatives are assumed to provide comparable monitoring service, an unverified functional-equivalence hypothesis. Maintenance emissions, replacement batteries, transport, disposal and facility utilization beyond the entered hours are excluded unless absorbed into the inputs. All amounts are CAD and undiscounted carbon masses are kg CO2e.

Assumption	Value
fleet	4.0000
robot_capex	250,000.0000
annual_opex	35,000.0000
hours	1,800.0000
robot_kw	0.8000
qml_kw	2.0000
qml_capex	100,000.0000
qml_embodied	5,000.0000
qml_reduction	0.0000
electricity_price	0.1500
grid_kg_kwh	0.0400
embodied_kg	12,000.0000
baseline_cost	250,000.0000
baseline_kg	25,000.0000
annual_benefit	100,000.0000

Assumption	Value
carbon_price	100.0000
discount	0.0400
prime	5.0000

Table 1. Exact tab input keys. qml_reduction is an assumed fractional robot-energy reduction (0 by default), not learned or experimentally demonstrated. qml_kw is total shared facility draw, including assumed cryogenic overhead; it is not per robot.

2. Finite economics and carbon derivations

$$C_0 = N\,c_{robot} + c_{QML};\ E = H\,[N\,P_{robot}(1-s) + P_{QML}]. \quad (1)$$

$$O = N\,o_{robot} + E\,p_{electricity};\ F = C_{baseline} + B - O. \quad (2)$$

$$NPV_T = -C_0 + \sum_{t=1}^{T} F/(1+r)^t,\ T = 10. \quad (3)$$

Power is kW, H is operating hours/year, and E is kWh/year. B is a hypothetical annual non-carbon benefit; it must not double-count savings already in the avoided baseline cost. NPV is undiscounted net cash flow when r = 0. Constant cash flows, no inflation, salvage or replacements are assumed.

$$G_{op} = E\,\gamma;\ G_{emb} = N\,g_{robot} + g_{QML};\ G_{project}(T) = G_{emb} + T\,G_{op}. \quad (4)$$

$$\Delta G(T) = T\,G_{baseline} - G_{project}(T);\ t_{carbon} = G_{emb}/(G_{baseline} - G_{op}). \quad (5)$$

The break-even ratio is meaningful only for positive annual avoided emissions. If annual emissions savings are nonpositive and embodied emissions are positive, no finite carbon break-even exists under this stationary model.

$$V_{carbon} = p_{CO2}(G_{baseline} - G_{op})/1000;\ NPV_{shadow} = -C_0 - p_{CO2}G_{emb}/1000 + \sum_{t=1}^{10}(F+V_{carbon})/(1+r)^t. \quad (6)$$

The carbon price is CAD/tonne CO2e. Carbon-valued NPV is a shadow-value scenario, not revenue or evidence of saleable offsets. Viability is defined here as financial NPV >= 0 and Delta G(10) >= 0, without claiming broader social viability.

3. Scenario results and sensitivity

Initial capital: 1,100,000.00 CAD.

Annual electricity: 9,360.00 kWh.

Annual operating cost: 141,404.00 CAD.

Annual operational emissions: 374.40 kg CO2e.

Embodied emissions: 53,000.00 kg CO2e.

Financial NPV: 591,900.42 CAD.

Carbon-valued NPV: 606,573.98 CAD.

Net avoided ten-year emissions: 193,256.00 kg CO2e.

Carbon break-even: 2.15 years.

Viable under the stated joint criterion: True.

Figure 1. Discounted value (CAD)

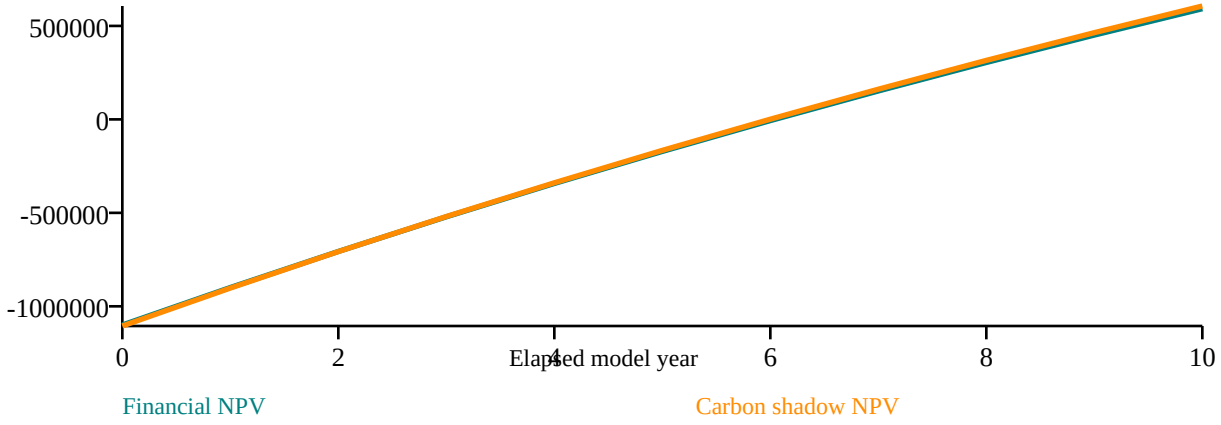
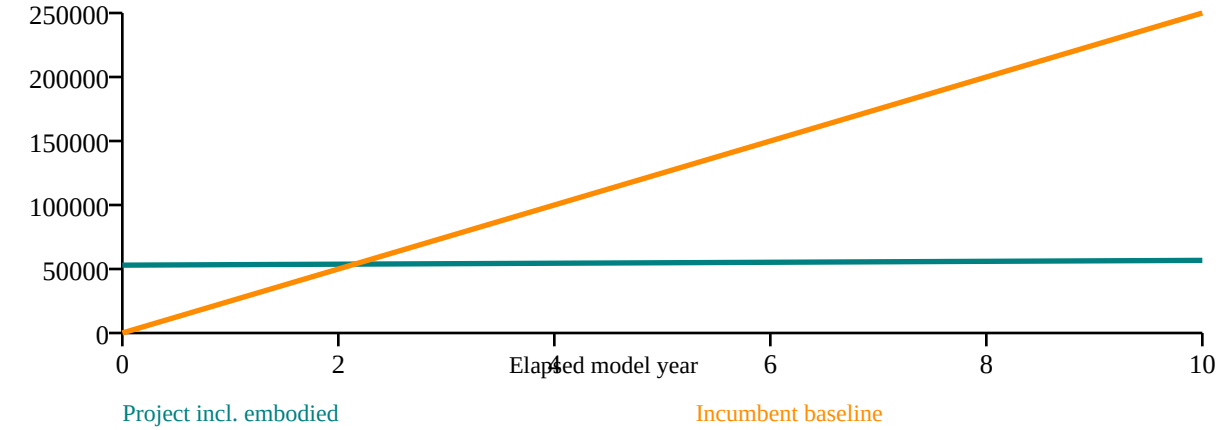


Figure 2. Lifecycle carbon (kg CO2e)



$$\mathrm{NPV}(\alpha C_0) = \mathrm{NPV}(C_0) - (\alpha - 1)C_0. \quad (7)$$

The tab sweeps capital multipliers 0.5, 0.75, 1, 1.25 and 1.5 while holding other parameters fixed. This is sensitivity analysis, not a probabilistic confidence interval.

4. Quantum kernel and number-theoretic structure

$$|\phi(x)\rangle = \bigotimes_{j=1}^{3} R_y(\pi x_j)|0\rangle; \; K(x,z) = \prod_{j=1}^{3} \cos^2[\pi(x_j - z_j)/2]. \quad (8)$$

$$\alpha = (K + 0.01\, I)^{-1} y; \; f(x) = k(x)^{T}\alpha. \quad (9)$$

Six synthetic training points have labels 0.6 x1 + 0.3 x2 + 0.1 x3. Query features are robot_kw/1000, hours/8760 and qml_reduction/0.5. The kernel is computed exactly on a classical computer, without entangling gates. Its training RMSE is 0.006090 and query score is 0.029885. Neither score is used in the financial forecast. Synthetic training accuracy is not evidence of useful QML prediction or speedup; real training data and held-out classical comparisons are required.

$$\text{group}(i) = i \bmod p;\ c_q(n) = \sum_{d \mid \gcd(q,n)} d\,\mu(q/d). \quad (10)$$

Prime-based residue groups label station schedules. The reused Privacy Cloak helper yields c5(1) = -1. These identities are finite arithmetic, not optimal geographic placement, encryption or a security guarantee. Lattice parameters belong to the standardized PQC provider, not to user-selected scheduling primes.

5. Post-quantum report verification

$$m = \text{domain} \,\|\, \text{canonicalJSON}(\text{report});\ \sigma = \text{Sign}_{sk}(m);\ \text{Verify}_{pk}(m,\sigma). \quad (11)$$

$$(ct,k_1) = \text{Encaps}(pk);\ k_2 = \text{Decaps}(sk,ct);\ \text{test: } k_1 = k_2 \text{ and } k_1 \neq \text{Decaps}(sk,\text{altered}(ct)). \quad (12)$$

The implementation uses pqcrypto 1.0.0 with ML-DSA-65 signatures and ML-KEM-768 shared-secret functional tests. RFC 8785 JSON canonicalization normalizes keys and numeric representations across browser round-trips, and rejects nonfinite numbers. A domain prefix binds reports to this app. The verifier pins the server's current public key instead of trusting a supplied key. Keys are ephemeral and reports expire on server restart.

signature_valid: **True**.

altered_report_rejected: **True**.

kem_shared_secret_agreed: **True**.

altered_ciphertext_secret_mismatch: **True**.

Public key fingerprint (SHA3-256): 68daa9668f2c1fa5ac6572bb6ee5b6ba265c5e7b8476da6bae981a4c749c60fc

These checks establish local functional behavior only. HTTP is not encrypted and the demo does not provide PQ TLS, user authentication, replay resistance, durable identity, encrypted reports or FIPS-certified module operation. The PDF itself is not digitally signed; its narrative records a report self-test.

6. Discussion, limitations and data availability

Economic viability depends strongly on comparable service, cryogenic overhead, manufacturing inventory, fleet utilization and attributed benefits. Entering measured values is supported, but source provenance is not independently checked. Monitoring is not equivalent to remediation or carbon removal. Cryptographic integrity cannot validate scientific assumptions, carbon claims or data quality. The source functions evaluate, quantum_kernel_demo and ReportSigner reproduce the calculations. No experiments or Nature affiliation are asserted.

References

NIST FIPS 203, Module-Lattice-Based Key-Encapsulation Mechanism Standard (2024), doi:10.6028/NIST.FIPS.203. NIST FIPS 204, Module-Lattice-Based Digital Signature Standard (2024), doi:10.6028/NIST.FIPS.204. Schuld and Killoran, Quantum Machine Learning in Feature Hilbert Spaces, Physical Review Letters 122, 040504 (2019), doi:10.1103/PhysRevLett.122.040504. These references motivate methods, not the numerical assumptions or security certification of this application.